2. **Assertion:** The Relative Molecular Mass of Chlorine is 35.5 a.m.u.

   Reason: The natural abundance of Chlorine isotopes are not equal.

VI. Short answer questions

1. Define: Relative atomic mass.
2. Write the different types of isotopes of oxygen and its percentage abundance.
3. Define: Atomicity
4. Give any two examples for heterodiatomic molecules.
5. What is Molar volume of a gas?
6. Find the percentage of nitrogen in ammonia.

VII. Long answer questions

1. Calculate the number of water molecule present in one drop of water which weighs 0.18 g.
2. $N_2 + 3\ H_2 \rightarrow 2\ NH_3$

   (The atomic mass of nitrogen is 14, and that of hydrogen is 1)

   1 mole of nitrogen (_____g) +
   3 moles of hydrogen (_____ g) →

   2 moles of ammonia (_____ g)
3. Calculate the number of moles in

   i) 27g of Al ii) 1.51×10^{23} molecules of NH_4Cl
4. Give the salient features of "Modern atomic theory".
5. Derive the relationship between Relative molecular mass and Vapour density.

VIII. HOT question

1. Calcium carbonate is decomposed on heating in the following reaction

   $$CaCO_3 \rightarrow CaO + CO_2$$

   i. How many moles of Calcium carbonate are involved in this reaction?

   ii. Calculate the gram molecular mass of calcium carbonate involved in this reaction

   iii. How many moles of CO_2 are there in this equation?

IX. Solve the following problems

1. How many grams are there in the following?

   i. 2 moles of hydrogen molecule, H_2

   ii. 3 moles of chlorine molecule, Cl_2

   iii. 5 moles of sulphur molecule, S_8

   iv. 4 moles of phosphorous molecule, P_4
2. Calculate the % of each element in calcium carbonate. (Atomic mass: C-12, O-16, Ca -40)
3. Calculate the % of oxygen in $Al_2(SO_4)_3$. (Atomic mass: Al-27, O-16, S -32)
4. Calculate the % relative abundance of B-10 and B-11, if its average atomic mass is 10.804 amu.

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INTERNET RESOURCES

https://www2.estrellamountain.edu/faculty/farabee/biobk/BioBookCHEM1.html

https://www.toppr.com/guides/chemistry/atoms-and-molecules/

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